

SOP-09: Thiol Reducing Agents and Alkylating Reagents (PHS - 2-Mercaptoethanol, DEPC)

Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 0 - September 1, 2026

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Room 209
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	_____ Date: _____
This SOP was created by (if not PI)	_____ Date: _____
Emergency contact	9-1-1 (campus phone) EH&S 206-543-7262 (M-F 8-5) 206-685-UWPD (8973) after hours
Date of last review	_____ Next review due: _____ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
2-Mercaptoethanol (beta-mercaptoethanol, BME), reagent grade	60-24-2
Diethyl pyrocarbonate (DEPC)	1609-47-8
2-Iodoacetamide	144-48-9
TCEP (tris(2-carboxyethyl)phosphine hydrochloride)	51805-45-9

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Skull and crossbones (2-mercaptoethanol, acute toxicity Cat 2 dermal and inhalation); Health hazard (reproductive toxicity, sensitization, STOT); Corrosion (TCEP, skin corrosion Cat 1B, eye damage Cat 1); Exclamation mark; Flame (combustible)

- PARTICULARLY HAZARDOUS SUBSTANCE. 2-Mercaptoethanol is classified GHS Acute toxicity Category 2 by both the dermal and inhalation routes - it is a substance of high acute toxicity - and Reproductive toxicity Category 2 (suspected of damaging fertility). DEPC is a suspected carcinogen and a reactive alkylating agent. 2-Iodoacetamide is an alkylating agent with Acute toxicity Category 3 (oral) and is a respiratory and skin sensitizer. Sections 9-11 of this SOP are mandatory.

- Stock chemicals: 2-mercaptoethanol (100 mL), DEPC (5 mL), 2-iodoacetamide (5 g solid), TCEP (2 g solid). Intermediates and working solutions: BME added to RNA lysis buffers at 1% v/v and to protein sample buffers at 5% v/v; DEPC added to water at 0.1% v/v and autoclaved to make nuclease-free water; iodoacetamide prepared as a fresh aqueous or buffer solution for cysteine alkylation in proteomics sample preparation; TCEP prepared as a neutralized aqueous stock for disulfide reduction. Wastes: BME-containing buffers and lysates, unreacted DEPC and DEPC-treated water before autoclaving, iodoacetamide solutions, and contaminated solid debris.
- Routes of exposure: inhalation of vapor (BME and DEPC), skin contact and absorption (BME, highly significant), eye contact, ingestion, and inhalation of dust when weighing iodoacetamide or TCEP powders.
- How exposure might occur: opening the BME bottle at the bench - the odor is detectable far below a hazardous concentration but the absence of odor is not proof of safety; pipetting BME into buffers; opening tubes of BME-containing lysate; pipetting DEPC into water; weighing iodoacetamide or TCEP powder; and skin contact with buffers containing these reagents.
- Target organs: 2-mercaptoethanol - central nervous system, respiratory tract, liver, kidney, eyes, skin, and the reproductive system. DEPC - respiratory tract, skin, and (as an alkylating agent) genetic material. 2-Iodoacetamide - respiratory tract and skin (sensitization) and, as an alkylating agent, all rapidly dividing tissue. TCEP - skin and eyes (corrosive).
- Signs and symptoms of exposure: BME - headache, dizziness, nausea, drowsiness, eye and airway irritation; at high concentrations, unconsciousness. Skin contact causes irritation and can deliver an acutely toxic dose. DEPC - eye, skin and respiratory irritation; DEPC hydrolyzes rapidly in water to ethanol and carbon dioxide, so a closed bottle of DEPC-treated water can build pressure and spray on opening. Iodoacetamide - skin rash, respiratory sensitization with wheeze on re-exposure. TCEP - immediate pain and burns on skin or eye contact.
- Expected by-products: DEPC hydrolysis produces ethanol and carbon dioxide (pressurization hazard); DEPC reacts with ammonia or Tris buffers, so DEPC is never added to Tris-containing solutions. BME oxidizes in air to a disulfide and has a persistent, penetrating odor that will be noticed throughout the building if a container is left open.
- GHS classification (from MyChem / SDS): 2-Mercaptoethanol - Acute toxicity Category 2 (dermal); Acute toxicity Category 2 (inhalation); Acute toxicity Category 3 (oral); Flammable liquids Category 4; Reproductive toxicity Category 2; Serious eye damage Category 1; Skin corrosion/irritation Category 2; Skin sensitization Category 1; STOT-repeated Category 2 (oral); STOT-single Category 3. DEPC - Acute toxicity Category 4 (oral); Flammable liquids Category 4; Combustible Liquid Class IIIA. 2-Iodoacetamide - Acute toxicity Category 3 (oral); Respiratory sensitization Category 1; Skin sensitization Category 1. TCEP - Serious eye damage Category 1; Skin corrosion Category 1B.
- Occupational exposure limits: no OSHA or WISHA PEL has been established for 2-mercaptoethanol, DEPC, iodoacetamide or TCEP. Control is by containment and work practice rather than by exposure limit.

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- ALL handling of 2-mercaptoethanol and DEPC - opening the bottle, pipetting, adding to buffers, and opening tubes of BME-containing lysate - is performed inside the certified chemical fume hood in FTR 209 with the sash at or below the marked working height. Any chemical fume hood used must be tested and passed by EH&S.
- Weighing 2-iodoacetamide and TCEP powders is performed inside the fume hood with the balance placed in the hood, on a disposable absorbent liner, to control dust.
- Once BME-containing buffer is diluted into a closed sample tube and the tube exterior is clean, closed-tube manipulation (vortexing, centrifuging, heating in a closed block) may be done at the bench; the tube is reopened only in the hood if the BME concentration is 1% or greater.
- DEPC-treated water bottles are autoclaved with the cap loosened to relieve pressure, and are cooled before the cap is tightened. Never tighten the cap on a bottle of freshly DEPC-treated water and leave it at room temperature.

- A disposable absorbent bench liner is used for all work under this SOP and discarded as hazardous solid waste.
- The nearest eyewash and safety shower are located in FTR 209 and are inspected and flushed weekly by lab staff.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.
- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: buttoned barrier lab coat. Because 2-mercaptoethanol is acutely toxic by skin absorption and TCEP is corrosive, additional protective clothing is REQUIRED where splashes or skin contact are foreseeable: disposable chemically resistant sleeves for all BME and DEPC work, and a chemically resistant apron plus face shield for any transfer of more than 10 mL of BME or for spill cleanup.
- Contaminated lab coats are removed immediately and bagged; BME-contaminated coats retain a strong odor and are discarded as hazardous waste rather than laundered.

Hand protection

- Hand protection IS required for the activities described in this SOP.
- Required gloves for 2-mercaptoethanol and DEPC: DOUBLE gloves - an inner 4-6 mil nitrile glove plus an outer butyl rubber or laminate (North Silver Shield) glove. Thin nitrile alone provides only brief protection against BME, which is acutely toxic dermally.
- Required gloves for weighing and handling 2-iodoacetamide and TCEP powders and their solutions: 6-8 mil nitrile, changed immediately on any contact.
- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with the specific chemical being used. Gloves must be inspected prior to use, including a check for pinholes.
- Any glove that contacts BME or DEPC is removed and replaced immediately, and the hand washed and inspected.
- Use proper glove removal technique (without touching the glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with these chemicals. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.
- Minimum eye protection: ANSI Z87.1 chemical splash goggles for all work with 2-mercaptoethanol, DEPC and TCEP, all of which cause serious eye damage. Safety glasses with side shields are acceptable for weighing iodoacetamide inside the hood and for closed-tube handling of dilute working solutions. A face shield over goggles is required for transfers of more than 10 mL of BME and for all spill cleanup.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP, provided all BME and DEPC handling is confined to the certified chemical fume hood as written.

- 2-Mercaptoethanol and DEPC are NEVER permitted to be used outside of the chemical fume hood. Closed tubes containing dilute BME buffer may be handled at the bench. Iodoacetamide and TCEP solutions, once prepared, may be used at the bench.
- The strong odor of BME is not a reliable exposure indicator and is not by itself evidence of an overexposure; conversely, working with BME outside a hood because 'the smell is not that bad' is prohibited.
- No lab member is currently enrolled in the EH&S Respiratory Protection Program for these chemicals.
- Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted), and when any Action Limit (AL) or Occupational Exposure Limit (OEL) has been exceeded or when there is a possibility that an AL/OEL will be exceeded. If a potential exposure hazard cannot be eliminated, contact the EH&S Respiratory Protection Program administrator to discuss respiratory protection or to enroll in the program so a respiratory protection analysis can be performed. Program enrollment includes medical evaluation, training and fit testing for an appropriate respirator.
- Where air-purifying respirators are appropriate, use a full-face respirator with organic vapor cartridges as a backup to engineering controls. Use a full-face supplied air respirator if it is the sole means of protection.

Section 4 - Special Handling and Storage Requirements

- Quantity limits: no more than 10 mL of neat 2-mercaptoethanol and no more than 1 mL of neat DEPC is dispensed at one time. Iodoacetamide and TCEP are weighed in quantities of 500 mg or less per operation. Larger operations require prior written PI approval.
- BME and DEPC bottles are opened only inside the fume hood and are recapped immediately after dispensing. A BME bottle left open at the bench will odorize the building and prompt a reported odor complaint; this has occurred in other UW buildings and is treated as a reportable near-miss here.
- DEPC is never added to Tris-containing buffers or to solutions containing amines or ammonia, with which it reacts. DEPC-treated water must be autoclaved to destroy residual DEPC before use in enzymatic reactions.
- 2-Iodoacetamide is light sensitive and is handled under reduced light; solutions are prepared fresh and are not stored.
- Because 2-iodoacetamide is a respiratory and skin sensitizer, any lab member who has previously reacted to it must not handle it, and the PI must be informed so that the task can be reassigned.
- All working solutions must be labeled with the full chemical name, concentration, the words 'TOXIC' (BME), 'SENSITIZER' (iodoacetamide) or 'CORROSIVE' (TCEP), the date and the preparer's initials.
- Work with neat BME or DEPC is performed only when at least one other person is present in or immediately adjacent to FTR 209. Unattended operations with open BME or DEPC are prohibited.
- A 'THIOL / ALKYLATING REAGENT IN USE - PARTICULARLY HAZARDOUS SUBSTANCE' sign is posted on the fume hood sash while work is in progress.
- Storage: 2-mercaptoethanol is stored in the toxics cabinet in FTR 209 in a secondary containment tray, tightly closed, away from oxidizers, acids and bases; because of its odor, the closed bottle is kept inside a sealed secondary container. DEPC is stored refrigerated per the manufacturer in a sealed secondary container, away from water, amines and bases. Iodoacetamide and TCEP solids are stored in the refrigerator or freezer as directed, in labeled, closed secondary containers, protected from light and moisture.
- Transport: containers travel in closed, leak-proof secondary containment and are not carried outside FTR 209.
- Clean the chemical fume hood upon completion of tasks with a detergent solution followed by clean water and then 70% ethanol; wipe dry. Residual BME odor in the hood indicates incomplete cleanup - repeat.
- Clean all contaminated surfaces with detergent solution followed by clean water and dry.
- Place all contaminated disposable items in appropriate laboratory waste for disposal, in a closed container - BME-contaminated tips left in an open trash can will odorize the room.

- Non-disposable/re-usable utensils, glassware, and other surfaces contaminated with these reagents must be decontaminated at the end of the laboratory work session. Complete this inside the chemical fume hood before removing any of the items.
- When work is completed, remove gloves and wash hands with soap and water.

Section 5 - Spill and Accident Procedures

Spill response

- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable. Evacuate the area and call 9-1-1 on a campus phone for help.
- Small spill - less than 5 mL of neat BME or DEPC contained INSIDE the operating fume hood, or less than 50 mL of dilute BME buffer: may be handled by trained lab staff. Leave the hood running with the sash low. Wearing goggles, face shield, butyl or laminate gloves over nitrile, lab coat with sleeves, cover the spill with universal absorbent pads, work from the outside inward, and place used absorbent directly into a screw-top waste container inside the hood. Then wash the area with detergent solution.
- Any spill of neat BME or DEPC greater than 5 mL, ANY spill of neat BME or DEPC outside the fume hood, or any spill in which a person may have been exposed: DO NOT clean up. Evacuate FTR 209, close the door, post a warning sign, prevent entry, and call 9-1-1 from a campus phone. Notify EH&S. Because BME is acutely toxic by inhalation, a bench spill in a small room is treated as an evacuation event, not a wipe-up.
- POWDER spill (2-iodoacetamide or TCEP): do not brush or wipe dry. Dampen with water inside the fume hood, wipe up with damp absorbent working from the outside inward, and place in a screw-top waste container. Any powder spill outside the fume hood is treated as a large spill - evacuate and call EH&S.
- PPE required for cleanup: chemical splash goggles AND face shield, butyl or laminate gloves over 6-8 mil nitrile, buttoned lab coat with chemically resistant apron and disposable sleeves, full-length pants and close-toe shoes.
- Signage and entry restrictions: post 'DO NOT ENTER - CHEMICAL SPILL' on the door for any spill outside the hood; restrict entry until EH&S clears the space. Be aware that responding personnel and neighboring labs will smell BME well beyond the affected room; the odor is not itself proof of a hazardous concentration, but the room is not reoccupied until EH&S clears it.
- Spill area must be cleaned up in the following manner: after absorbent removal, clean the spill area thoroughly with detergent solution followed by clean water, then 70% ethanol, and dry. A dilute bleach wipe may be used on BME residue on non-metal surfaces to oxidize the thiol and reduce odor, but ONLY after all liquid has been absorbed and removed and never where guanidinium-containing material may be present.
- Spill cleanup materials must be disposed of in the following manner: double bag all absorbent and contaminated disposables, place in a screw-top container inside the fume hood, label with a UW Hazardous Waste Label naming the reagent and 'spill debris', and submit a collection request to EH&S promptly.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours (Monday-Friday, 8 a.m. to 5 p.m.).
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours (certain types of incidents require immediate notification).

Exposure procedures - first aid

- Perform first aid immediately. Refer to the SDS for additional chemical-specific guidance.
- Skin exposure to 2-mercaptoethanol is a medical emergency because BME is acutely toxic by the dermal route. Immediately remove contaminated gloves and clothing and use the nearest safety shower for at least 15 minutes; stay

under the shower while removing clothing; use a clean lab coat or spare clothing for cover-up. Call 9-1-1 for any contact larger than a small splash.

- Skin exposure to DEPC, iodoacetamide or TCEP: wash with soap and running water for 15 minutes, using the safety shower for larger contact; seek medical evaluation for any burn (TCEP) or rash (iodoacetamide).
- Eye exposure: use the eyewash for 15 minutes while holding the eyelids open, then go immediately to the Emergency Department. BME and TCEP cause serious eye damage.
- Inhalation exposure: move the person out of the contaminated area to fresh air immediately; get medical help. For BME, call 9-1-1 - inhalation acute toxicity is Category 2.
- Ingestion: rinse mouth with water; do not induce vomiting; call 9-1-1.
- Sharps injury (needle stick or subcutaneous exposure): scrub the exposed area thoroughly for 15 minutes using warm water and sudsing soap; seek immediate medical evaluation.
- Get help. Call 9-1-1 or go to the nearest Emergency Department; provide details of exposure: agent (name the reagent; for BME state that it is acutely toxic by skin and inhalation), dose, route of exposure, and time since exposure. Bring the SDS and this SOP to the Emergency Department.
- Notify your supervisor as soon as possible for assistance. Secure the area before leaving; lock doors and indicate the spill if needed.
- Notify EH&S immediately after providing first aid and/or getting help. During business hours (M-F, 8-5) call 206-543-7262. Outside of business hours call 206-685-UWPD (8973) to be routed to EH&S staff on call.
- Any person who develops a rash, wheeze or eye and airway irritation after working with 2-iodoacetamide should report it to the PI and EH&S - these are signs of sensitization, and further exposure must be avoided.

Section 6 - Waste Accumulation and Disposal Procedures

- Waste streams: (a) neat 2-mercaptoethanol and BME-containing buffers and lysates; (b) neat DEPC and untreated DEPC water; (c) iodoacetamide solutions and unused solid; (d) TCEP solutions and unused solid; (e) contaminated solid debris (tips, tubes, liners, gloves, weigh boats, absorbent).
- Do NOT pour any of these reagents down the drain. Do NOT autoclave BME- or iodoacetamide-containing waste. DEPC-treated water that has been fully autoclaved contains only ethanol and carbon dioxide breakdown products and may be handled as non-hazardous aqueous waste; untreated DEPC solution may not.
- Keep these wastes separate from acid waste, oxidizer waste, and guanidinium-containing extraction waste.
- Accumulate liquid waste at the point of generation - inside the fume hood - in a sturdy, chemically compatible HDPE container with a securely closable screw-top lid, kept closed except when actively adding waste, in secondary containment. Because of BME odor, keep the waste container inside the hood at all times and consider a secondary sealed bag.
- Never seal a container that may still be evolving gas; DEPC hydrolysis releases carbon dioxide and can pressurize a closed waste container. Vent DEPC waste inside the hood before capping and check the container periodically.
- All chemical waste containers must be labeled with a UW Hazardous Waste Label listing every constituent and approximate percentage, starting from the moment the first drop of waste is added. Refer to 'How to Label Chemical Waste Containers' on the EH&S website.
- Solid debris is collected in a lined, closed, labeled solid hazardous waste container kept inside or immediately adjacent to the fume hood.
- Manage chemical and hazardous chemical waste separately from other waste streams such as biohazardous waste. Never autoclave chemical waste because it can produce hazardous chemical vapors, aerosols, and explosive reactions.

- The University of Washington Environmental Health & Safety Department has full responsibility for collection of hazardous waste for the University, all its campuses, and off-site locations; University laboratories cannot contract with an outside vendor to collect hazardous waste.
- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions.
- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplusing materials. Update the MyChem inventory when containers are emptied, received, transferred or surplused.

Section 7 - Protocol

- Procedure A - Addition of 2-mercaptoethanol to lysis or sample buffer. Inside the certified fume hood, on an absorbent liner, wearing goggles, double gloves, lab coat and sleeves: open the BME bottle, withdraw the required volume with a pipette (typically 10 microliters per mL of buffer for 1% v/v), add it to the buffer in a capped-ready tube or bottle, and recap the BME bottle immediately. Cap and invert the buffer to mix. Wipe the exterior of the buffer container before removing it from the hood. Discard the tip into the closed solid waste container inside the hood.
- Procedure B - Preparation of DEPC-treated water. Inside the fume hood, add DEPC to deionized water to 0.1% v/v in a bottle with the cap loose, mix, and let stand overnight at room temperature or 37 degC inside the hood. Autoclave with the cap loose to destroy residual DEPC and vent carbon dioxide. Cool before tightening the cap. Label with 'DEPC-treated, autoclaved', the date and initials. Never add DEPC to Tris buffer.
- Procedure C - Weighing 2-iodoacetamide or TCEP. Inside the fume hood, with the balance in the hood and an absorbent liner in place, wearing goggles, nitrile gloves and a lab coat: weigh the solid into a disposable weigh boat, transfer to a labeled tube, dissolve in the specified buffer inside the hood, and cap. Prepare iodoacetamide solutions fresh and protect from light. Discard the weigh boat, spatula and liner into the closed solid waste container.
- Procedure D - Use of working solutions. Dilute BME buffers in closed tubes, iodoacetamide alkylation reactions and TCEP reduction reactions may be carried out at the bench in capped tubes with nitrile gloves, goggles and a lab coat. Open tubes containing 1% or more BME only inside the fume hood.
- Prohibited: opening neat BME or DEPC outside the fume hood; adding DEPC to Tris; tightly capping freshly DEPC-treated water; autoclaving BME or iodoacetamide waste; drain disposal; use by untrained or unauthorized personnel.
- Refer to Section 2 of the UW Laboratory Safety Manual on the EH&S website for additional guidance on chemical management and preparation for use for particularly hazardous substances (PHSs).
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

No. N/A - none of these reagents are administered to animals under this SOP.

IACUC protocol number (if applicable): _____

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 9 - Approvals Required

- All staff working with the reagents covered by this SOP must be trained on this SOP prior to starting work. They must also review the SDS for each reagent, and the SDSs must be readily available in the laboratory (MyChem). All training must be documented and maintained by the PI or their designee.
- 2-Mercaptoethanol is a particularly hazardous substance (acute toxicity Category 2 dermal and inhalation; reproductive toxicity Category 2). Written authorization from the Principal Investigator is required before any individual may work with neat 2-mercaptoethanol or neat DEPC, and that authorization is recorded on the training roster in Section 12 with the PI's countersignature.
- Prerequisite training, all documented before independent work: (1) EH&S Laboratory Safety Training, initial and current refresher; (2) EH&S Managing Laboratory Chemical Waste training; (3) documented review of this SOP and the SDSs; (4) documented instruction on glove selection for BME and on the DEPC pressurization hazard.
- A worker must complete at least one supervised run of Procedure A and, if applicable, Procedure B, observed and signed off by the PI or a designated experienced lab member, before working with neat BME or DEPC independently.
- Personnel who are pregnant, planning a pregnancy, or breastfeeding should consult the EH&S Reproductive Health in the Workplace resources and their healthcare provider and are encouraged to discuss alternative task assignment with the PI, because 2-mercaptoethanol is a suspected reproductive toxicant. Such consultation is voluntary and confidential.
- Any individual who has experienced a skin or respiratory reaction to 2-iodoacetamide must be reassigned away from that task; sensitized individuals can react at very low exposures.
- No medical examination is required for the work described because respirators are not used.

Section 10 - Decontamination

- Fume hood: at the end of every session, remove and bag the absorbent bench liner as hazardous solid waste, then wipe the hood work surface and inside of the sash with detergent solution, then clean water, then 70% ethanol; allow to dry with the hood running. Persistent BME odor means residue remains - repeat the detergent wash.
- Equipment: wipe pipettors, racks and the balance with detergent solution then 70% ethanol. Balance pans used for iodoacetamide or TCEP are wiped with a damp wipe and then dried; do not brush powder residue into the air.
- Glassware: rinse three times inside the fume hood, collecting the first rinse in the appropriate waste container, then wash with detergent and water and rinse three times with deionized water.
- Heat blocks and centrifuges used with closed BME tubes are wiped with detergent solution and 70% ethanol after use; a leaked tube requires decontamination of the block or rotor inside the hood.
- Controlled areas: there are no glove boxes or restricted-access hoods in FTR 209. The designated area in Section 11 is wiped down as described above at the end of each session and monthly regardless of use.
- Cleaning solutions and materials: laboratory detergent (Alconox or Contrad) solution, clean water, 70% ethanol, lint-free wipes, disposable absorbent bench liners.

Section 11 - Designated Area

- DESIGNATED AREA: the certified chemical fume hood in FTR 209, with a disposable absorbent liner and the balance placed inside it for powder weighing, is the sole designated area for all handling of neat 2-mercaptoethanol, neat DEPC, and solid 2-iodoacetamide and TCEP.
- The designated area is posted with a 'DESIGNATED AREA - PARTICULARLY HAZARDOUS SUBSTANCE - 2-MERCAPTOETHANOL / DEPC / ALKYLATING AGENTS' sign listing the PI, contact number, required PPE and the authorization requirement.
- Waste containers for these reagents are kept within the designated area, inside the fume hood, in secondary containment.

- Storage: the toxics cabinet in FTR 209 (BME, in a sealed secondary container) and the laboratory refrigerator/freezer in FTR 209 (DEPC, iodoacetamide, TCEP, each in a labeled closed secondary container); all storage positions are posted with the PHS designation.
- Closed-tube handling of dilute working solutions is permitted at the molecular biology bench in FTR 209 with the PPE specified in Section 3. No other bench, room or piece of equipment is authorized for open handling of these reagents.

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.
- The Principal Investigator, or Responsible Party if the activity does not involve a PI, must ensure that their laboratory personnel have attended appropriate laboratory safety training (and refresher training where applicable).
- Training must be repeated following any revision to the content of this SOP.
- Training must be documented. This training sheet is provided as one option; other forms of training documentation (including electronic) are acceptable but records must be accessible and immediately available upon request.
- Because this SOP covers a particularly hazardous substance, the Principal Investigator countersigns each entry below to record the written authorization required by Section 9.

I have read and understand the content of this SOP:

[illegible]